AI-Powered Auto-Responder

CIS 91 · Week 7 Midterm Project · Tat Thang Huynh (TH) · 2026-11-10 20:06:12.174 UTC

GitHub repository: `https://github.com/thuynh073/cis91-automation-workflows`
n8n workflow: `https://n8n-cis091.codislabs.club/workflow/clOIbUt0ShYirBXC`
Production webhook: `https://n8n-cis091.codislabs.club/webhook/midterm-auto-responder`

4	12	15	2	5/5
categories	knowledge base entries	workflow nodes	LLM calls per message	tests passed

1. Purpose

This system is an AI-powered auto-responder for a fictional campus IT help desk. A student or staff member sends a JSON message containing a sender, subject, and message body. The workflow classifies the message, retrieves relevant help desk knowledge, drafts a professional reply, returns that reply to the caller, and records the complete interaction. It is intended for high-volume first-line questions about VPN, Wi-Fi, passwords, MFA, the student portal, printing, software, tuition plans, refunds, transcripts, and instructor contact.

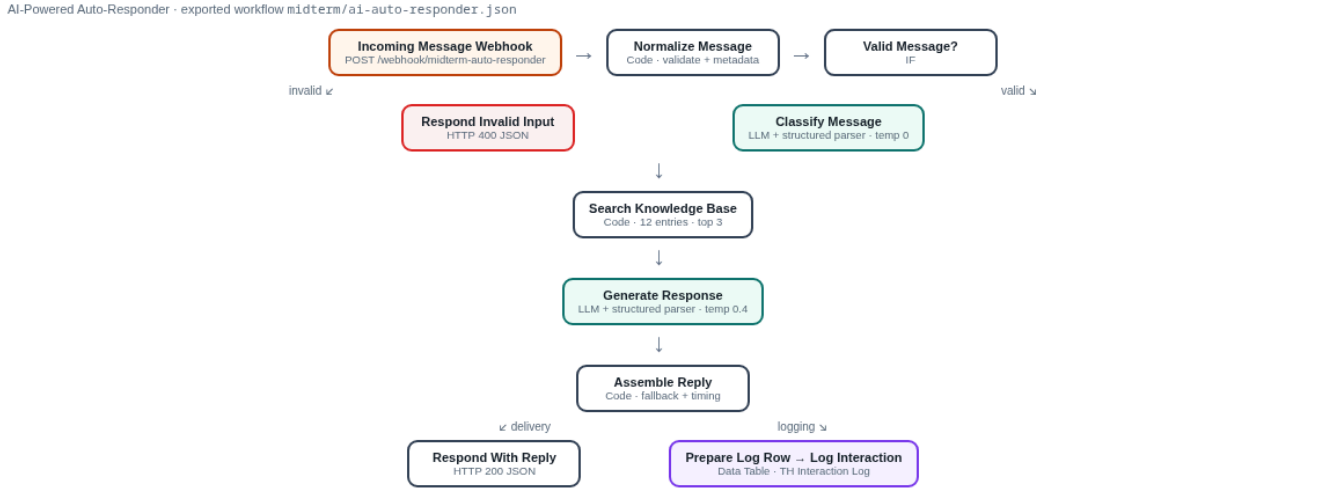
The design goal is not to replace a human, but to answer documented questions immediately and route anything ambiguous, urgent, or outside scope to a specialist with enough context to act.

2. Architecture

> **Design choice:** a webhook trigger is the easiest surface to test and to connect to a form, chat widget, or email parser later. Deterministic keyword retrieval was chosen over embeddings because it needs no vector database, runs instantly, and every match can be explained in the log. The LLM is used for language tasks (classification and drafting), not for deciding policy facts.

1. **Incoming Message Webhook** accepts `POST /webhook/midterm-auto-responder`. The JSON body arrives under `body`.
2. **Normalize Message** accepts `sender/from/email`, `subject/topic`, and `message/body/text/content` aliases, trims values, creates a message ID and timestamp, and validates that message text exists.
3. **Valid Message?** sends empty or malformed input to **Respond Invalid Input** (HTTP 400 with a JSON error). Valid input continues.
4. **Classify Message** is a LangChain LLM chain with a structured output parser. It returns `category`, `confidence`, `keywords`, and `reasoning` at temperature 0.
5. **Search Knowledge Base** is a Code node. It tokenizes the subject, body, and classifier keywords, removes stopwords, applies a light stemmer, and scores all 12 entries with IDF-weighted term overlap. Scores are saturated so a long message cannot win by length alone; the classifier's category adds a small tie-break bonus. The top three entries above 0.28 are kept, and a context block is built for the LLM.
6. **Generate Response** is a second LLM chain at temperature 0.4. It receives the original message, the classification, and the KB context, and returns `subject`, `body`, `used_kb_ids`, `escalation_needed`, and `tone`.
7. **Assemble Reply** applies a safe fallback if the model returned no body, calculates response time, and builds both the caller-facing reply and the log row.
8. **Respond With Reply** returns HTTP 200 JSON. In parallel, **Prepare Log Row** and **Log Interaction** write the interaction to the **TH Interaction Log** Data Table.

2. Architecture — workflow figure



3. Knowledge Base

The theme is a **campus IT help desk**. The knowledge base contains **12 entries**, each with an ID, category, canonical question, answer, and search keywords. Entries are embedded as a JSON array inside the Search Knowledge Base Code node, so the workflow is self-contained and needs no external sheet. Categories: technical_support (KB-01, 02, 04, 05, 10), faq (KB-03, 08, 09), billing (KB-06, 07), and general_inquiry (KB-11, 12). Every entry also carries search keywords; the full objects are in the workflow JSON.

KB-01 TECHNICAL_SUPPORT
How do I connect to the campus VPN from my laptop?

KB-02 TECHNICAL_SUPPORT
Why can't I get on the campus Wi-Fi?

KB-03 FAQ
How do I reset my campus password?

KB-04 TECHNICAL_SUPPORT
How do I set up Duo multi-factor authentication on a new phone?

KB-05 TECHNICAL_SUPPORT
The student portal or Canvas will not let me log in.

KB-06 BILLING
Does the college offer a tuition payment plan?

KB-07 BILLING
What is the refund policy if I drop a class?

KB-08 FAQ
What software can students install for free?

KB-09 FAQ
How do I print on campus?

KB-10 TECHNICAL_SUPPORT
My account is locked out after too many failed logins.

KB-11 GENERAL_INQUIRY
How do I request an official transcript?

KB-12 GENERAL_INQUIRY
How do I contact my instructor or find office hours?

4. AI Prompts

Classification prompt (temperature 0)

```
You are the triage classifier for the CIS 91 Help Desk, which answers
questions from students and staff about campus IT, accounts, and billing.

Classify the message below into exactly ONE of these categories:

- faq: a general "how do I..." question that the help desk documents cover
  (printing, software availability, hours, common how-tos).
- technical_support: something is broken or unreachable for this user -
  VPN, campus Wi-Fi, logins, MFA, locked accounts, portal errors.
- billing: money matters - tuition, payment plans, refunds, charges, holds,
  late fees, receipts.
- general_inquiry: anything that is not IT or money - transcripts,
  enrollment questions, contacting an instructor, office hours, policies.

Sender: {{ $json.sender }}
Subject: {{ $json.subject }}
Message:
{{ $json.original_message }}

Rules:
- Choose the category that matches the sender's primary goal, even if they
  mention several topics.
- confidence is a number from 0.0 to 1.0 and must reflect genuine ambiguity.
  Use below 0.6 when the message could reasonably fit two categories.
- keywords: 3 to 6 single lowercase words taken from the message, used to
  search the help desk knowledge base.
- reasoning: one or two sentences explaining the choice.
```

Response generation prompt (temperature 0.4)

```
You are a support agent for the CIS 91 Help Desk replying to a student
or staff member. Write the actual reply that will be sent to them.

Incoming message
Sender: {{ $json.sender }}
Subject: {{ $json.subject }}
Category chosen by triage: {{ $json.classified_category }} (confidence {{ $json.classified_confidence }})
Message:
{{ $json.original_message }}

Knowledge base matches ({{ $json.kb_match_type }} match)
{{ $json.kb_context }}

How to write the reply:
- Answer the sender's actual question first, in the first sentence.
- When knowledge base entries are provided, use their concrete details only -
  never invent policies, prices, phone numbers, or deadlines that are not in
  the knowledge base.
- If the match is "none" or the question is not covered, say plainly that you
  do not have that information and that you are routing the message to a
  specialist, then set escalation_needed to true.
- Set escalation_needed to true for billing disputes, account lockouts,
  lost devices, refunds outside policy, or anything with a deadline in the
  next 48 hours.
- Keep the body under 140 words, professional and warm, plain text only (no
  markdown headings or bullet characters).
- End with one clear next step.

Return subject (a short reply subject line), body, used_kb_ids (the KB ids you
actually used, or an empty array), escalation_needed (true or false), and tone
(one word, e.g. reassuring, apologetic, neutral).
```

Prompt design choices. The classification prompt defines each category with concrete examples and tells the model to classify by the sender's primary goal, not by every topic mentioned. It asks for a calibrated confidence (below 0.6 when two categories fit), 3–6 lowercase keywords for retrieval, and a short reason that can be audited in the log. Temperature 0 makes repeated messages consistent.

The response prompt puts the sender's actual question first, supplies only the matched KB context, and explicitly forbids inventing policies, prices, phone numbers, or deadlines. It forces a plain-text reply under 140 words, one clear next step, and an explicit escalation flag for billing disputes, account lockouts, lost devices, out-of-policy refunds, or deadlines within 48 hours. The no-match instruction is what prevents the weather/currency test from producing a fabricated answer. Temperature 0.4 keeps the wording natural without changing facts.

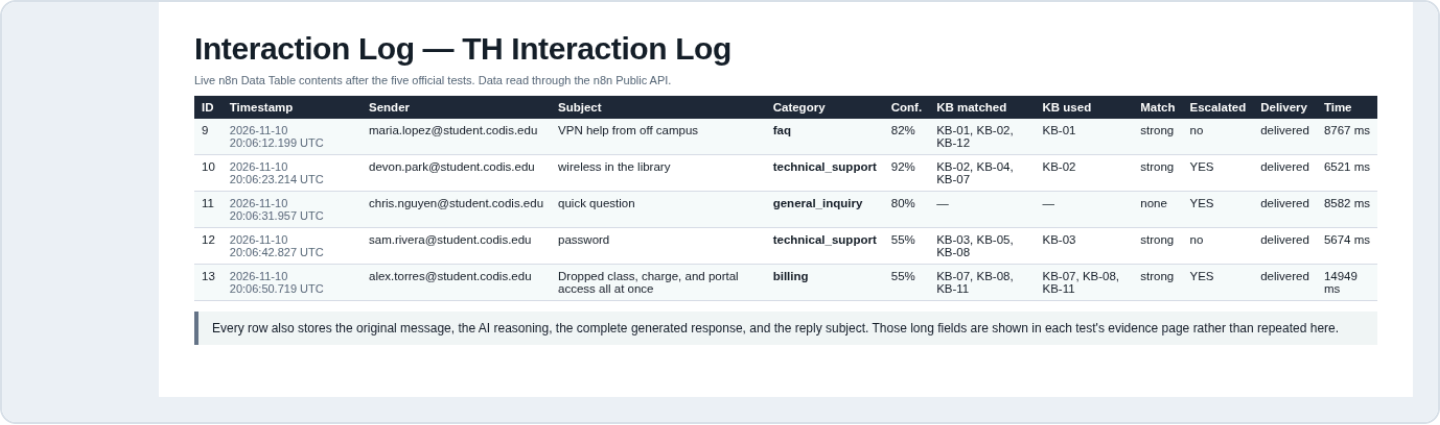
5. Limitations

- **Lexical retrieval, not semantic.** A paraphrase that shares no vocabulary with an entry can be missed. Embeddings/vector search would improve recall.
- **One label, and sometimes an ambiguous one.** The long test is classified billing even though it also contains a portal outage, a transcript deadline, and a software question; the top-three KB context compensates, but the category is lossy. "Password reset?" can be faq or technical_support, so the workflow logs 0.55 confidence instead of pretending certainty.
- **No webhook authentication, rate limiting, or deduplication.** A production deployment needs a shared secret or gateway, and should key log rows on a sender-provided message ID.
- **Delivery is webhook + Data Table only.** There is no SMTP/email send, no outbound retry queue, and no acknowledgement beyond the HTTP response.
- **Limited knowledge base and retention.** 12 entries cover the demo topics only, and the Data Table has no retention policy for message text that may contain personal data.
- **Latency, cost, and generic fallback.** Tests took roughly 6–15 seconds because each message makes two LLM calls. If both retries fail, the caller gets a safe escalation message rather than a specific answer, and no operator alert is sent.

Testing — Five Diverse Messages

The five cases below were sent to the production webhook with `curl` and the JSON files in `midterm/tests/`. Each evidence page is a screenshot of the captured n8n execution: the input, classification, KB matches, generated reply, and the resulting interaction-log row. All five executions returned HTTP 200 and logged with delivery status `delivered`.

Test	Type	Message (short)	Category · confidence	KB matched	Escalated	Delivery
Test 1	Direct match	How do I connect to the campus VPN from my laptop? I'm home for the week and need to reach the	faq · 82%	KB-01, KB-02, KB-12	no	delivered
Test 2	Partial match	I can't get the internet working on my phone in the library and my class starts in ten minutes.	technical_support · 92%	KB-02, KB-04, KB-07	yes	delivered
Test 3	No match	What is the weather going to be like in Paris this weekend, and can you also convert 50 US doll	general_inquiry · 80%	none	yes	delivered
Test 4	Short message	Password reset?	technical_support · 55%	KB-03, KB-05, KB-08	no	delivered
Test 5	Long message	Hi, I have a few things going on and I'm not sure who to ask. First, I dropped my statistics cl	billing · 55%	KB-07, KB-08, KB-11	yes	delivered



Interaction Log — TH Interaction Log

Live n8n Data Table contents after the five official tests. Data read through the n8n Public API.

ID	Timestamp	Sender	Subject	Category	Conf.	KB matched	KB used	Match	Escalated	Delivery	Time
9	2026-11-10 20:06:12.199 UTC	maria.lopez@student.codis.edu	VPN help from off campus	**faq**	82%	KB-01, KB-02, KB-12	KB-01	strong	no	delivered	8767 ms
10	2026-11-10 20:06:23.214 UTC	devon.park@student.codis.edu	wireless in the library	**technical_support**	92%	KB-02, KB-04, KB-07	KB-02	strong	YES	delivered	6521 ms
11	2026-11-10 20:06:31.957 UTC	chris.nguyen@student.codis.edu	quick question	**general_inquiry**	80%	—	—	none	YES	delivered	8582 ms
12	2026-11-10 20:06:42.827 UTC	sam.rivera@student.codis.edu	password	**technical_support**	55%	KB-03, KB-05, KB-08	KB-03	strong	no	delivered	5674 ms
13	2026-11-10 20:06:50.719 UTC	alex.torres@student.codis.edu	Dropped class, charge, and portal access all at once	**billing**	55%	KB-07, KB-08, KB-11	KB-07, KB-08, KB-11	strong	YES	delivered	14949 ms

> Every row also stores the original message, the AI reasoning, the complete generated response, and the reply subject. Those long fields are shown in each test's evidence page rather than repeated here.

The screenshot above is the live TH Interaction Log after all five tests. The per-test pages that follow show the full original message, reasoning and AI response stored in each row.

Test 1 — Direct match

Execution 2685 · 2026-11-10 20:06:12.174 UTC

PASS — correct answer, no escalation

Test 1 — Direct match

Execution 2685 · 2026-11-10 20:06:12.174 UTC

PASS — correct answer, no escalation

1 · INPUT MESSAGE

Sender	maria.lopez@student.codis.edu
Subject	VPN help from off campus
Message ID	`MSG-1794341172199-l75fk`

> How do I connect to the campus VPN from my laptop? I'm home for the week and need to reach the library databases for my research paper.

2 · AI CLASSIFICATION

Category	**faq**
Confidence	82%
Keywords	vpn, connect, campus, laptop, library, databases

Reasoning: The sender is asking a general how-to question about connecting to the campus VPN from off campus, not reporting a broken or unreachable service. This matches the help desk's FAQ category for common setup instructions.

3 · KNOWLEDGE BASE MATCHES

ID	Score	Question	Role
KB-01	0.96	How do I connect to the campus VPN from my laptop?	used
KB-02	0.72	Why can't I get on the campus Wi-Fi?	candidate
KB-12	0.63	How do I contact my instructor or find office hours?	candidate

4 · GENERATED RESPONSE

Subject	Re: VPN help from off campus
Tone	reassuring
Escalation	no
KB used	KB-01

> To connect from your laptop, install the GlobalProtect client from software.codis.edu/vpn, then sign in with your campus username and password and approve the Duo prompt. The gateway is vpn.codis.edu. When the client shows Connected, you can reach the library databases from off campus. If installation or sign-in gives you trouble, try the steps once more from a fresh browser session and confirm you're using your campus username rather than an email alias. Your next step: visit software.codis.edu/vpn now and install GlobalProtect before your research session.

5 · LOG ENTRY (TH INTERACTION LOG)

Row ID	9
Timestamp	2026-11-10 20:06:12.199 UTC
Category	faq · 82%
KB matched	KB-01, KB-02, KB-12
KB used	KB-01
Match type	strong (top 0.96)
Delivery	delivered via webhook
Priority	normal
Response time	8767 ms
Reply subject	Re: VPN help from off campus

Original message, reasoning, AI response and response status are all stored in the same row.

What happened: The message asks exactly how to connect to the campus VPN from a laptop. Triage labeled it **faq** (0.82) because the intent is a how-to, even though KB-01 is tagged technical_support. Search matched KB-01 (0.96), KB-02 (0.72), and KB-12 (0.63); the generated reply used KB-01 and included GlobalProtect, software.codis.edu/vpn, the vpn.codis.edu gateway, and the Duo prompt. The answer is accurate and complete.

What happened: The message asks exactly how to connect to the campus VPN from a laptop. Triage labeled it **faq** (0.82) because the intent is a how-to, even though KB-01 is tagged technical_support. Search matched KB-01 (0.96), KB-02 (0.72), and KB-12 (0.63); the generated reply used KB-01 and included GlobalProtect, software.codis.edu/vpn, the vpn.codis.edu gateway, and the Duo prompt. The answer is accurate and complete.

Test 2 — Partial match

Execution 2687 · 2026-11-10 20:06:23.190 UTC

PASS — correct answer, escalated for urgency

Test 2 — Partial match

Execution 2687 · 2026-11-10 20:06:23.190 UTC

PASS — correct answer, escalated for urgency

1 · INPUT MESSAGE

Sender	devon.park@student.codis.edu
Subject	wireless in the library
Message ID	`MSG-1794341183214-o710t`

> I can't get the internet working on my phone in the library and my class starts in ten minutes. Is there a different network I should be joining?

2 · AI CLASSIFICATION

Category	**technical_support**
Confidence	92%
Keywords	internet, phone, library, network, joining

Reasoning: The sender cannot get internet working on their phone in the library, which is a connectivity issue needing technical support. The question about joining a different network is part of troubleshooting the same problem.

3 · KNOWLEDGE BASE MATCHES

ID	Score	Question	Role
KB-02	1.00	Why can't I get on the campus Wi-Fi?	used
KB-04	0.68	How do I set up Duo multi-factor authentication on a new phone?	candidate
KB-07	0.63	What is the refund policy if I drop a class?	candidate

4 · GENERATED RESPONSE

Subject	Re: wireless in the library
Tone	reassuring
Escalation	YES
KB used	KB-02

> Join the CODID-Students network using your campus username (not your email alias) and password; if it keeps failing, forget the network first and then rejoin. If you need quick internet before class, CODID-Guest is open for 24 hours, but it does not reach internal servers. Because your class starts in ten minutes, I'm flagging this for a specialist in case you need further help. Please try rejoining CODID-Students now and reply if it still fails.

5 · LOG ENTRY (TH INTERACTION LOG)

Row ID	10
Timestamp	2026-11-10 20:06:23.214 UTC
Category	technical_support · 92%
KB matched	KB-02, KB-04, KB-07
KB used	KB-02
Match type	strong (top 1)
Delivery	delivered via webhook
Priority	high
Response time	6521 ms
Reply subject	Re: wireless in the library

Original message, reasoning, AI response and response status are all stored in the same row.

What happened: There is no exact VPN question; the sender describes phone internet in the library. Search still matched KB-02 at 1.00 through internet/phone/library/network, with KB-04 and KB-07 as weaker candidates. The reply used KB-02 and gave the CODID-Students and CODID-Guest instructions. It also set escalation_needed=true because the class starts in ten minutes, which is the right call.

What happened: There is no exact VPN question; the sender describes phone internet in the library. Search still matched KB-02 at 1.00 through internet/phone/library/network, with KB-04 and KB-07 as weaker candidates. The reply used KB-02 and gave the CODID-Students and CODID-Guest instructions. It also set escalation_needed=true because the class starts in ten minutes, which is the right call.

Test 3 — No match

Execution 2688 · 2026-11-10 20:06:31.935 UTC

PASS — no invented answer, routed to a specialist

Test 3 — No match

Execution 2688 · 2026-11-10 20:06:31.935 UTC

PASS — no invented answer, routed to a specialist

1 · INPUT MESSAGE

Sender	chris.nguyen@student.codis.edu
Subject	quick question
Message ID	`MSG-1794341191957-gqiw1`

> What is the weather going to be like in Paris this weekend, and can you also convert 50 US dollars into euros for me?

2 · AI CLASSIFICATION

Category	**general_inquiry**
Confidence	80%
Keywords	weather, paris, weekend, convert, dollars, euros

Reasoning: The message asks about weather and a personal currency conversion, neither of which is a campus IT issue or a student billing matter. It is best treated as a general inquiry outside the help desk's documented categories.

3 · KNOWLEDGE BASE MATCHES

ID	Score	Question	Role
No knowledge base entry matched this message.			

4 · GENERATED RESPONSE

Subject	Re: quick question
Tone	neutral
Escalation	YES
KB used	—

> I don't have access to live weather forecasts or currency conversion rates, so I can't confirm the Paris weather this weekend or convert 50 US dollars into euros for you. I'm routing your message to a specialist who can help with this request. Please watch for their follow-up.

5 · LOG ENTRY (TH INTERACTION LOG)

Row ID	11
Timestamp	2026-11-10 20:06:31.957 UTC
Category	general_inquiry · 80%
KB matched	—
KB used	—
Match type	none (top 0)
Delivery	delivered via webhook
Priority	high
Response time	8582 ms
Reply subject	Re: quick question

Original message, reasoning, AI response and response status are all stored in the same row.

What happened: Weather in Paris and a USD-to-EUR conversion are outside the knowledge base. Triage returned **general_inquiry** (0.80); the search returned zero entries. The model stated plainly that it cannot access live forecasts or exchange rates, did not invent data, and routed the message to a specialist with escalation_needed=true.

What happened: Weather in Paris and a USD-to-EUR conversion are outside the knowledge base. Triage returned **general_inquiry** (0.80); the search returned zero entries. The model stated plainly that it cannot access live forecasts or exchange rates, did not invent data, and routed the message to a specialist with escalation_needed=true.

Test 4 — Short message

Execution 2689 · 2026-11-10 20:06:42.807 UTC

PASS — correct content; category is ambiguous

Test 4 — Short message

Execution 2689 · 2026-11-10 20:06:42.807 UTC

PASS — correct content; category is ambiguous

1 · INPUT MESSAGE

Sender	sam.rivera@student.codis.edu
Subject	password
Message ID	MSG-1794341202827-waeic

> Password reset?

2 · AI CLASSIFICATION

Category	**technical_support**
Confidence	55%
Keywords	password, reset, student

Reasoning: The message asks about resetting a password, which is a login/account access issue. It could also be a general how-to FAQ, so confidence is moderate.

3 · KNOWLEDGE BASE MATCHES

ID	Score	Question	Role
KB-03	0.86	How do I reset my campus password?	used
KB-05	0.59	The student portal or Canvas will not let me log in.	candidate
KB-08	0.54	What software can students install for free?	candidate

4 · GENERATED RESPONSE

Subject	Re: password reset
Tone	reassuring
Escalation	no
KB used	KB-03

> You can reset your campus password at accounts.codis.edu/reset. Enter your campus username, confirm with Duo, and set a new password of at least 12 characters. The change should reach Wi-Fi, VPN, and the portal within about 5 minutes. Please visit accounts.codis.edu/reset now to complete your reset.

5 · LOG ENTRY (TH INTERACTION LOG)

Row ID	12
Timestamp	2026-11-10 20:06:42.827 UTC
Category	technical_support · 55%
KB matched	KB-03, KB-05, KB-08
KB used	KB-03
Match type	strong (top 0.86)
Delivery	delivered via webhook
Priority	normal
Response time	5674 ms
Reply subject	Re: password reset

Original message, reasoning, AI response and response status are all stored in the same row.

What happened: The terse message 'Password reset?' was classified **technical_support** (0.55). Search matched KB-03 (0.86), KB-05 (0.59), and KB-08 (0.54), and the reply used KB-03. The reply gives the reset URL, the 12-character rule, and the five-minute propagation window. The content is correct, but the category could reasonably be faq; the low confidence reflects that ambiguity.

What happened: The terse message 'Password reset?' was classified **technical_support** (0.55). Search matched KB-03 (0.86), KB-05 (0.59), and KB-08 (0.54), and the reply used KB-03. The reply gives the reset URL, the 12-character rule, and the five-minute propagation window. The content is correct, but the category could reasonably be faq; the low confidence reflects that ambiguity.

Test 5 — Long message

Execution 2690 · 2026-11-10 20:06:50.699 UTC

PASS — all three KB topics addressed, urgent items escalated

Test 5 — Long message

Execution 2690 · 2026-11-10 20:06:50.699 UTC

PASS — all three KB topics addressed, urgent items escalated

1 · INPUT MESSAGE

Sender	alex.torres@student.codis.edu
Subject	Dropped class, charge, and portal access all at once
Message ID	`MSG-1794341210719-mcysw`

> Hi, I have a few things going on and I'm not sure who to ask. First, I dropped my statistics class in week three because of a scheduling conflict with my new job, and I still see the full tuition charge on my account. I thought dropping in week three got me at least part of the money back. Second, ever since I dropped it, the student portal keeps kicking me back to the login screen with an error code and I can't get into Canvas to check my other grades, so I also need my official transcript request to go through before a scholarship deadline on Friday. Finally, could you tell me what software I can install for free on my own laptop, because the lab closes at 9pm and I do most of my work after that? Sorry for the long message, I just wanted to get everything into one email.

2 · AI CLASSIFICATION

Category	**billing**
Confidence	55%
Keywords	tuition, charge, dropped, portal, login

Reasoning: The message mentions several issues, but the first and most specific concern is the full tuition charge after dropping a class and a possible refund, which is a billing matter. The portal/login error is a significant competing issue, so confidence is moderate.

3 · KNOWLEDGE BASE MATCHES

ID	Score	Question	Role
KB-07	1.00	What is the refund policy if I drop a class?	used
KB-08	0.98	What software can students install for free?	used
KB-11	0.98	How do I request an official transcript?	used

4 · GENERATED RESPONSE

Subject	Re: Dropped class, charge, portal, and software
Tone	reassuring
Escalation	YES
KB used	KB-07, KB-08, KB-11

> Since you dropped in week three, you should receive a 50% refund; refunds return to the original payment method within 10 business days. I'm escalating the portal login error and your Friday transcript deadline to a specialist because both need account access. Once the portal works, order your official transcript through the National Student Clearinghouse under Records > Transcripts: electronic is $8 and usually arrives within one business day, paper is $12 and takes 5-7 business days. For free software, use software.codis.edu with your campus account to get Microsoft 365, JetBrains, GitHub Student Developer Pack, and VMware. Next step: please reply with the portal error code so I can attach it to the escalation.

5 · LOG ENTRY (TH INTERACTION LOG)

Row ID	13
Timestamp	2026-11-10 20:06:50.719 UTC
Category	billing · 55%
KB matched	KB-07, KB-08, KB-11
KB used	KB-07, KB-08, KB-11
Match type	strong (top 1)
Delivery	delivered via webhook
Priority	high
Response time	14949 ms
Reply subject	Re: Dropped class, charge, portal, and software

Original message, reasoning, AI response and response status are all stored in the same row.

What happened: The multi-paragraph message covers a week-three drop and charge, a portal login error, a Friday transcript deadline, and free software. Triage chose **billing** (0.55) because the refund is the first and most specific concern. Search matched KB-07, KB-08, and KB-11 at 0.98–1.00, and the reply used all three: the 50% week-three refund, transcript ordering, and the free software list. It escalated the portal error and the Friday deadline. A single category cannot represent all four topics, which is a documented limitation.

What happened: The multi-paragraph message covers a week-three drop and charge, a portal login error, a Friday transcript deadline, and free software. Triage chose **billing** (0.55) because the refund is the first and most specific concern. Search matched KB-07, KB-08, and KB-11 at 0.98–1.00, and the reply used all three: the 50% week-three refund, transcript ordering, and the free software list. It escalated the portal error and the Friday deadline. A single category cannot represent all four topics, which is a documented limitation.